Telco Customer Churn

In [127…

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder,PowerTransformer,OrdinalEncoder,C
import seaborn as sns
from sklearn.model_selection import train_test_split
from matplotlib import pyplot as plt
import numpy as np
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from imblearn.over_sampling import SMOTE
from sklearn.impute import SimpleImputer
from sklearn.metrics import precision_score, recall_score, confusion_matrix, Con
from sklearn.model_selection import RandomizedSearchCV
```

In [117…

```
df = pd.read_csv('https://raw.githubusercontent.com/KarthiObli/TelcoCustomerChur
```

In [117…

```
df.head()
```

Out[117…

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns

In [117…

```
df.drop('customerID',axis=1,inplace=True)
```

In [117…

```
df_corr = df[:]
```

In [117…

```
le = LabelEncoder()
```

In [117…

```
for i in df_corr.select_dtypes(include='object').columns:
  df_corr[i] = le.fit_transform(df_corr[i])
```

In [117…

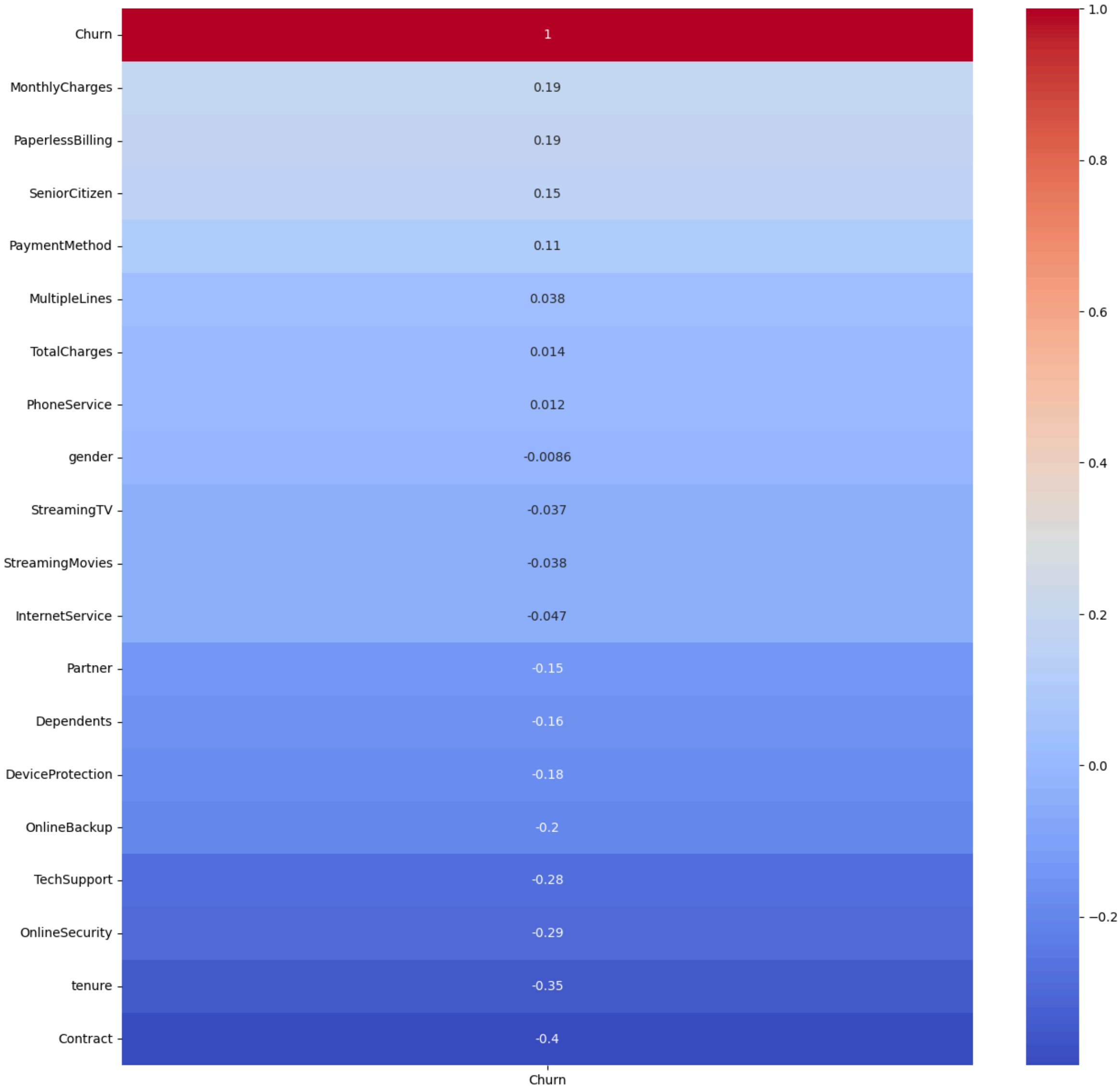
```
corr = df_corr.corr()
```

In [118…

```
plt.figure(figsize=(15,15))
sns.heatmap(corr[['Churn']].sort_values(by='Churn',ascending=False),annot=True,c
```

Out[118…

```
<Axes: >
```



In [118…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   gender            7043 non-null   object 
 1   SeniorCitizen     7043 non-null   int64  
 2   Partner           7043 non-null   object 
 3   Dependents        7043 non-null   object 
 4   tenure            7043 non-null   int64  
 5   PhoneService      7043 non-null   object 
 6   MultipleLines     7043 non-null   object 
 7   InternetService   7043 non-null   object 
 8   OnlineSecurity    7043 non-null   object 
 9   OnlineBackup      7043 non-null   object 
 10  DeviceProtection  7043 non-null   object 
 11  TechSupport       7043 non-null   object 
 12  StreamingTV       7043 non-null   object 
 13  StreamingMovies   7043 non-null   object 
 14  Contract          7043 non-null   object 
 15  PaperlessBilling  7043 non-null   object 
 16  PaymentMethod     7043 non-null   object 
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7043 non-null   object 
 19  Churn             7043 non-null   object 
dtypes: float64(1), int64(2), object(17)
memory usage: 1.1+ MB
```

In [118…
```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'].str.strip(),errors='coerce
```

In [118…
```
df['TotalCharges'] = df['TotalCharges'].astype('float')
```

In [118…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   gender            7043 non-null   object 
 1   SeniorCitizen     7043 non-null   int64  
 2   Partner           7043 non-null   object 
 3   Dependents        7043 non-null   object 
 4   tenure            7043 non-null   int64  
 5   PhoneService      7043 non-null   object 
 6   MultipleLines     7043 non-null   object 
 7   InternetService   7043 non-null   object 
 8   OnlineSecurity    7043 non-null   object 
 9   OnlineBackup      7043 non-null   object 
 10  DeviceProtection  7043 non-null   object 
 11  TechSupport       7043 non-null   object 
 12  StreamingTV       7043 non-null   object 
 13  StreamingMovies   7043 non-null   object 
 14  Contract          7043 non-null   object 
 15  PaperlessBilling  7043 non-null   object 
 16  PaymentMethod     7043 non-null   object 
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7032 non-null   float64
 19  Churn             7043 non-null   object 
dtypes: float64(2), int64(2), object(16)
memory usage: 1.1+ MB
```

In [118…

```python
df.describe()
```

Out[118…

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7032.000000
mean	0.162147	32.371149	64.761692	2283.300441
std	0.368612	24.559481	30.090047	2266.771362
min	0.000000	0.000000	18.250000	18.800000
25%	0.000000	9.000000	35.500000	401.450000
50%	0.000000	29.000000	70.350000	1397.475000
75%	0.000000	55.000000	89.850000	3794.737500
max	1.000000	72.000000	118.750000	8684.800000

In [118…

```python
df['SeniorCitizen'] = np.where(df['SeniorCitizen']==1,'Yes','No')
```

Train Test Split

In [118…

```python
x = df.drop('Churn',axis=1)
```

In [118…

```python
y = df[['Churn']]
```

In [118…

```python
x_train, x_test, y_train, y_test = train_test_split(x,y,random_state=21,test_siz
```

In [119…

```
x_train.reset_index(drop=True,inplace=True)
x_test.reset_index(drop=True,inplace=True)
y_train.reset_index(drop=True,inplace=True)
y_test.reset_index(drop=True,inplace=True)
```

Null Handling

In [119…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   object
 1   SeniorCitizen     7043 non-null   object
 2   Partner           7043 non-null   object
 3   Dependents        7043 non-null   object
 4   tenure            7043 non-null   int64
 5   PhoneService      7043 non-null   object
 6   MultipleLines     7043 non-null   object
 7   InternetService   7043 non-null   object
 8   OnlineSecurity    7043 non-null   object
 9   OnlineBackup      7043 non-null   object
 10  DeviceProtection  7043 non-null   object
 11  TechSupport       7043 non-null   object
 12  StreamingTV       7043 non-null   object
 13  StreamingMovies   7043 non-null   object
 14  Contract          7043 non-null   object
 15  PaperlessBilling  7043 non-null   object
 16  PaymentMethod     7043 non-null   object
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7032 non-null   float64
 19  Churn             7043 non-null   object
dtypes: float64(2), int64(1), object(17)
memory usage: 1.1+ MB
```

In [119…

```
Impute_Total_Charges = SimpleImputer(strategy = 'mean')
x_train['TotalCharges'] = Impute_Total_Charges.fit_transform(x_train[['TotalChar
x_test['TotalCharges'] = Impute_Total_Charges.fit_transform(x_test[['TotalCharge
```

Outlier Handling

In [119…

```
x_train.head()
```

Out[119...

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	In
0	Male	No	No	No	2	Yes	Yes	
1	Female	No	No	No	34	Yes	Yes	
2	Female	No	No	No	9	Yes	No	
3	Male	No	Yes	No	67	Yes	Yes	
4	Male	No	No	No	71	No	No phone service	

In [119...

```
numerical_columns = [i for i in x_train.select_dtypes(['int','float']).columns]
```

In [119...

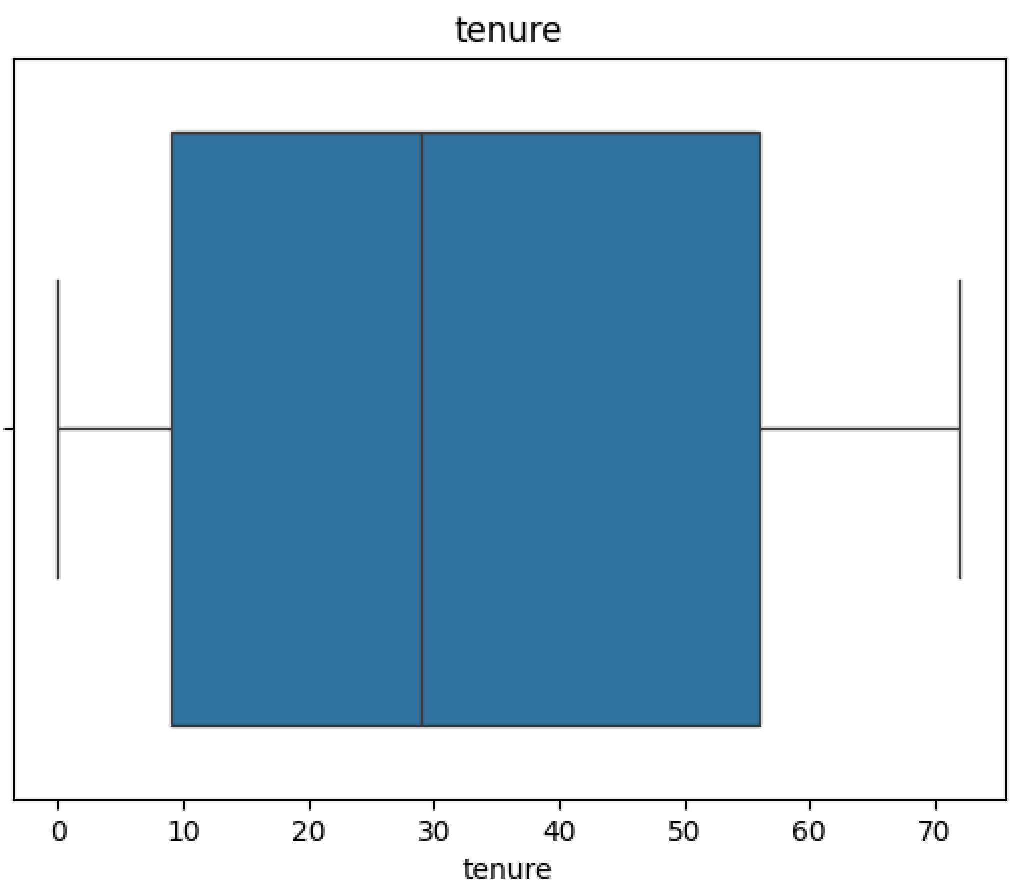
```
numerical_columns
```

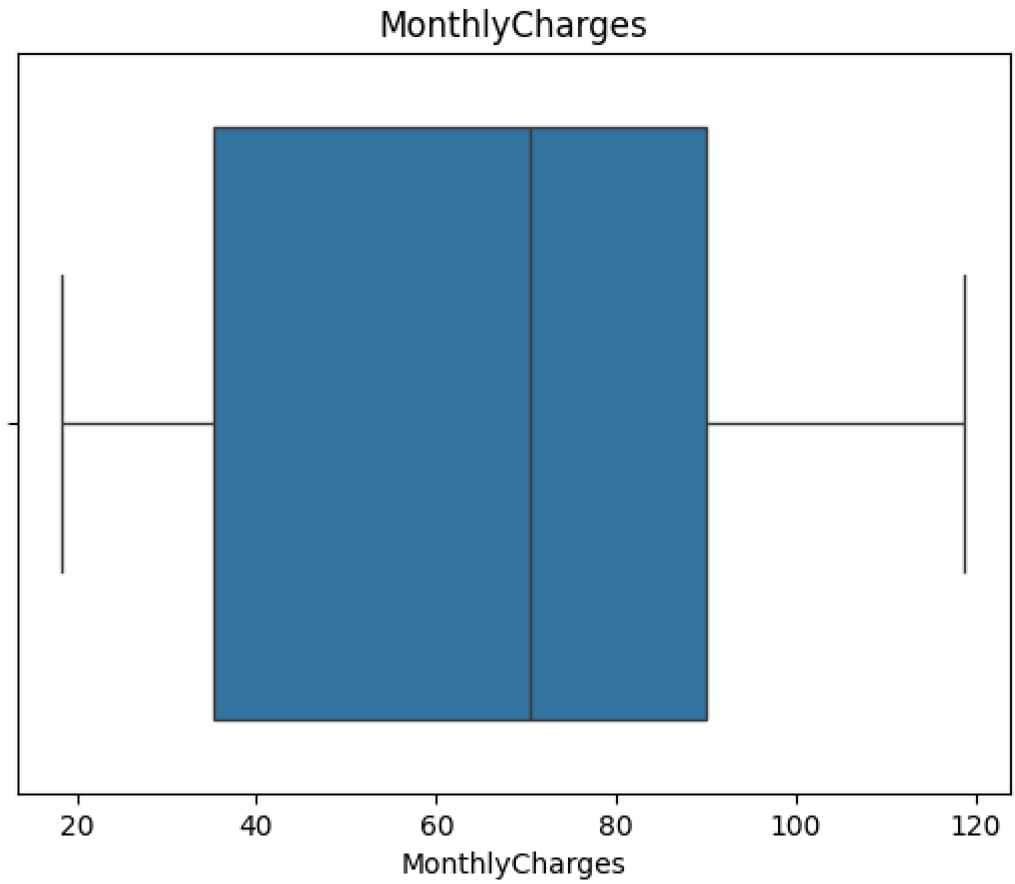
Out[119...

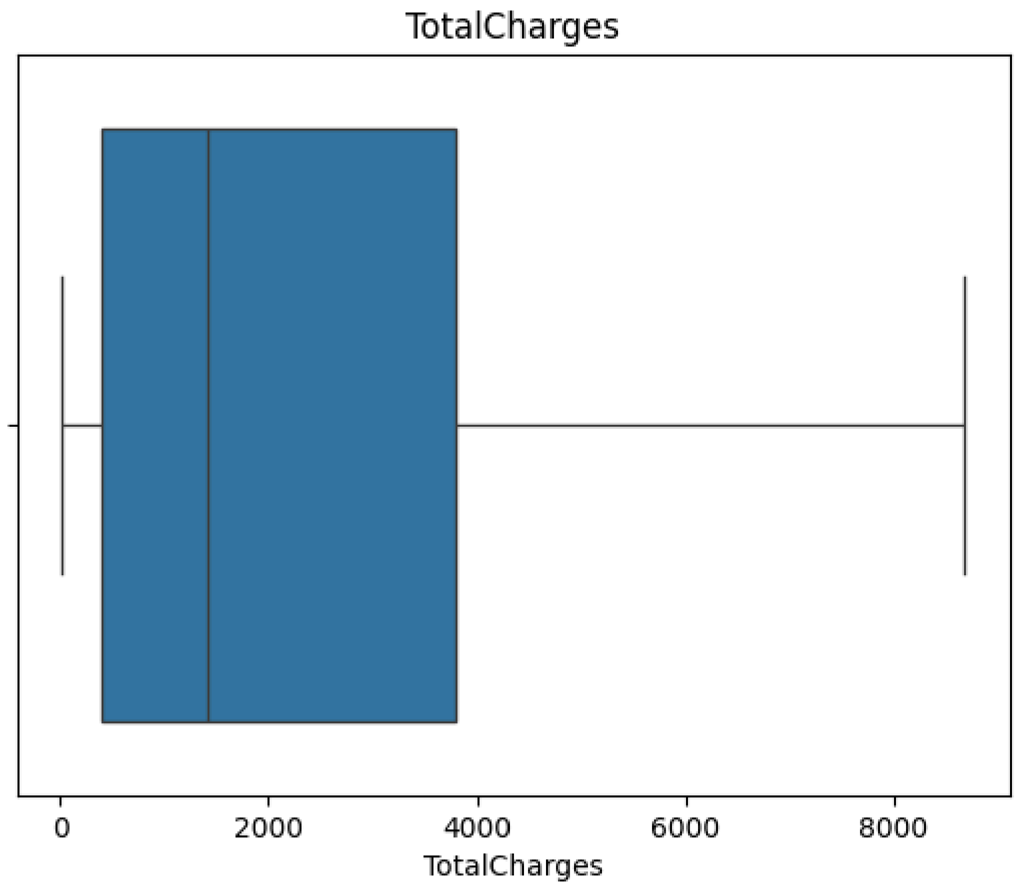
```
['tenure', 'MonthlyCharges', 'TotalCharges']
```

In [119...

```
for i in numerical_columns:
  sns.boxplot(x=x_train[i])
  plt.title(i)
  plt.show()
```







Outlier handling is not required

Skew Data handling

In [119…

```
for i in numerical_columns:
  sns.distplot(x=x_train[i])
  plt.title(i)
  plt.show()
```

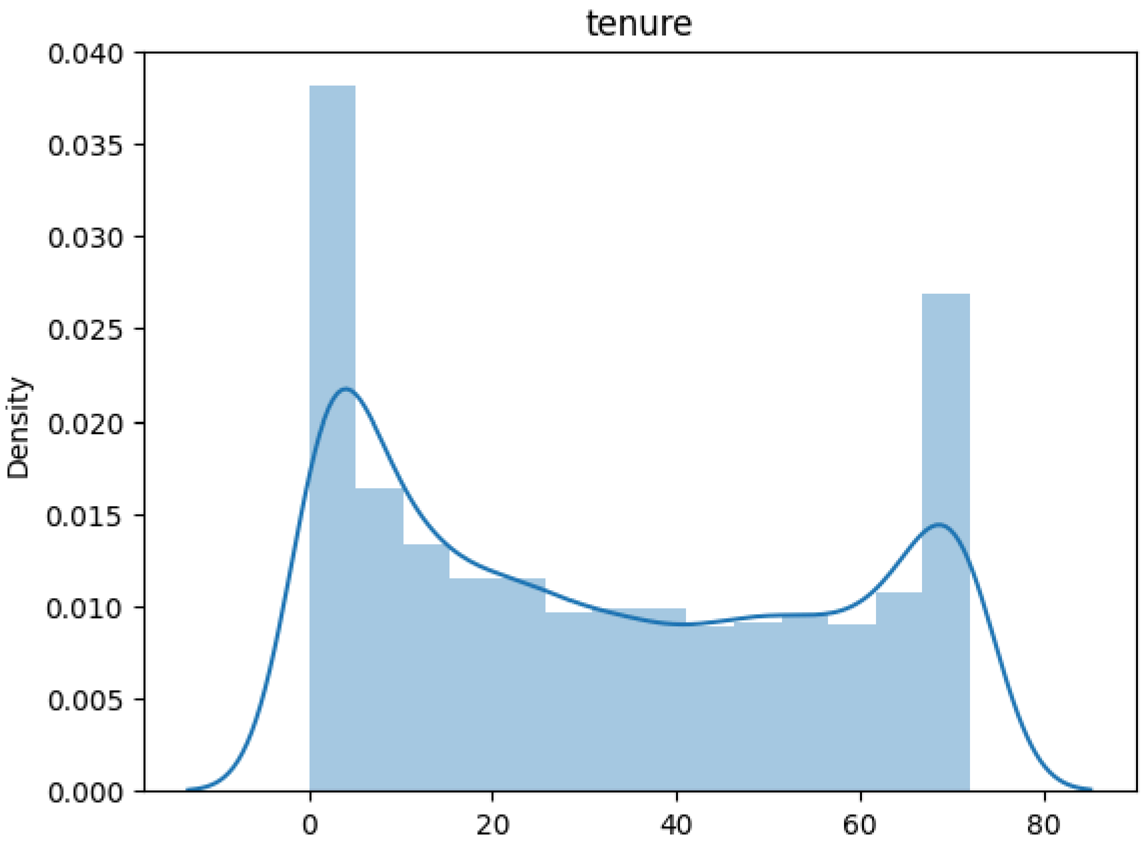
```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x=x_train[i])
```




```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x=x_train[i])
```

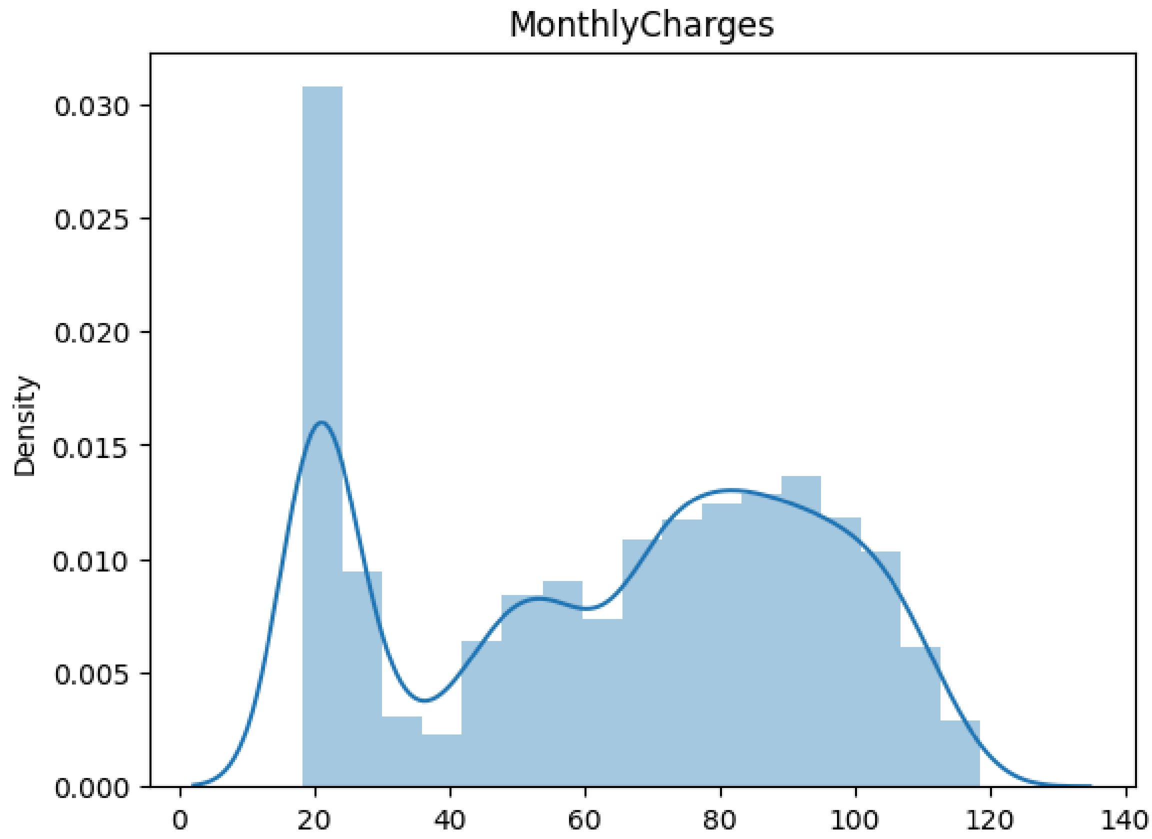


```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

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  sns.distplot(x=x_train[i])
```
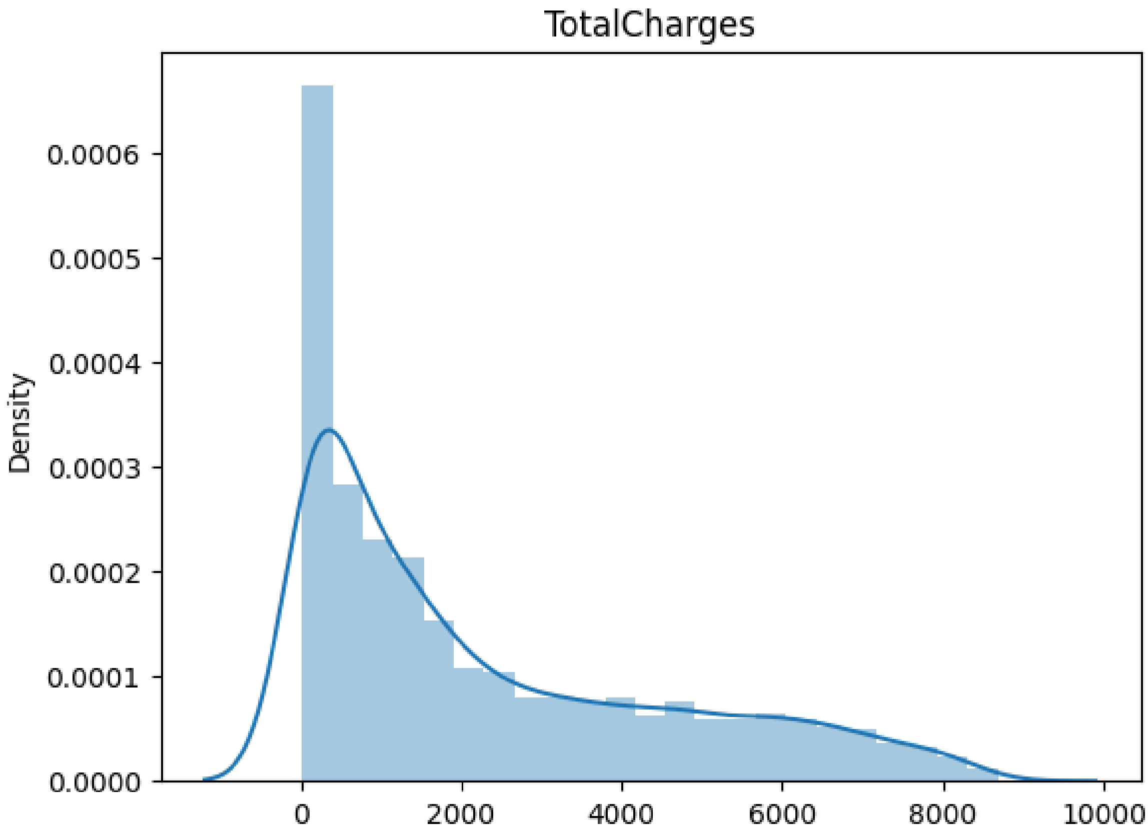



In [119…
```
for i in numerical_columns:
  print(f'Skew of {i} is {x_train[i].skew()}')
```

```
Skew of tenure is 0.24154307949177942
Skew of MonthlyCharges is -0.2180067984814727
Skew of TotalCharges is 0.959567310717357
```

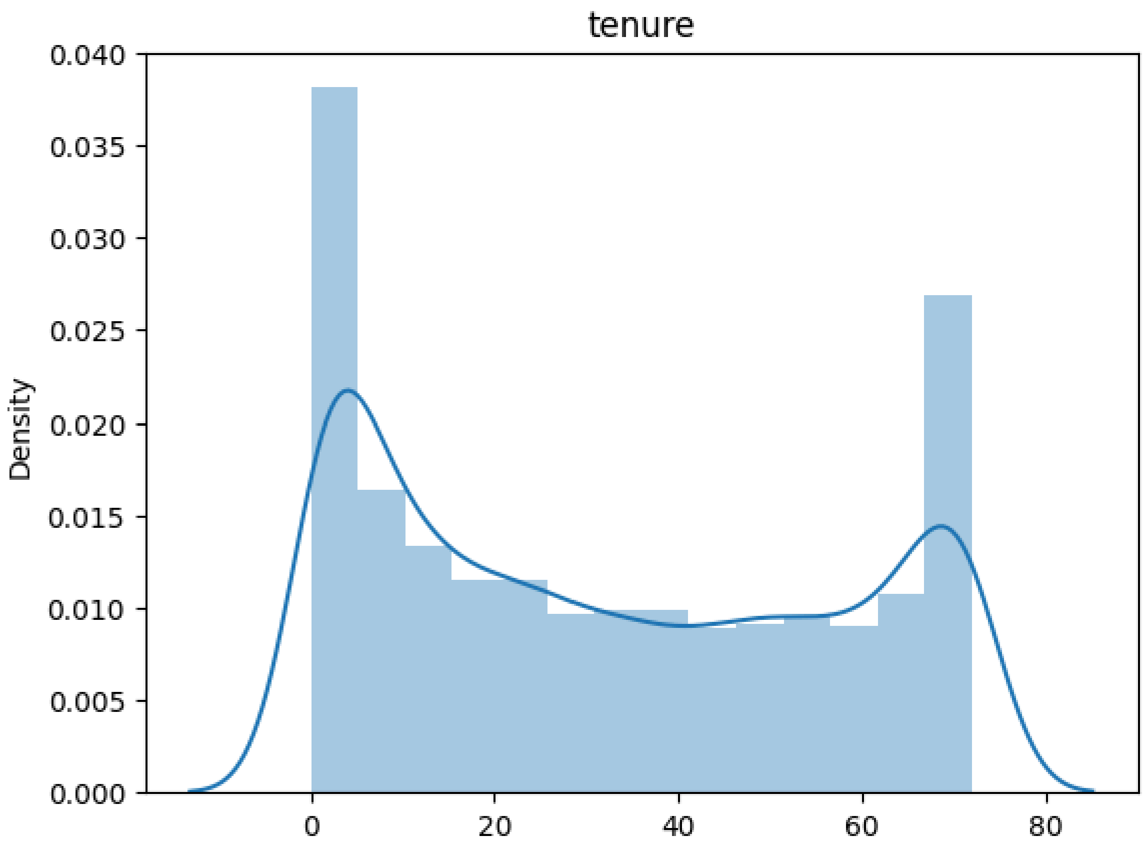
In [119…
```
PT = PowerTransformer(method='yeo-johnson')
```

In [120…
```
x_train['TotalCharges'] = PT.fit_transform(x_train[['TotalCharges']])
```

In [120…
```
x_test['TotalCharges'] = PT.transform(x_test[['TotalCharges']])
```

In [120…
```
for i in numerical_columns:
  sns.distplot(x=x_train[i])
  plt.title(i)
  plt.show()
```

```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x=x_train[i])
```




```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
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For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x=x_train[i])
```
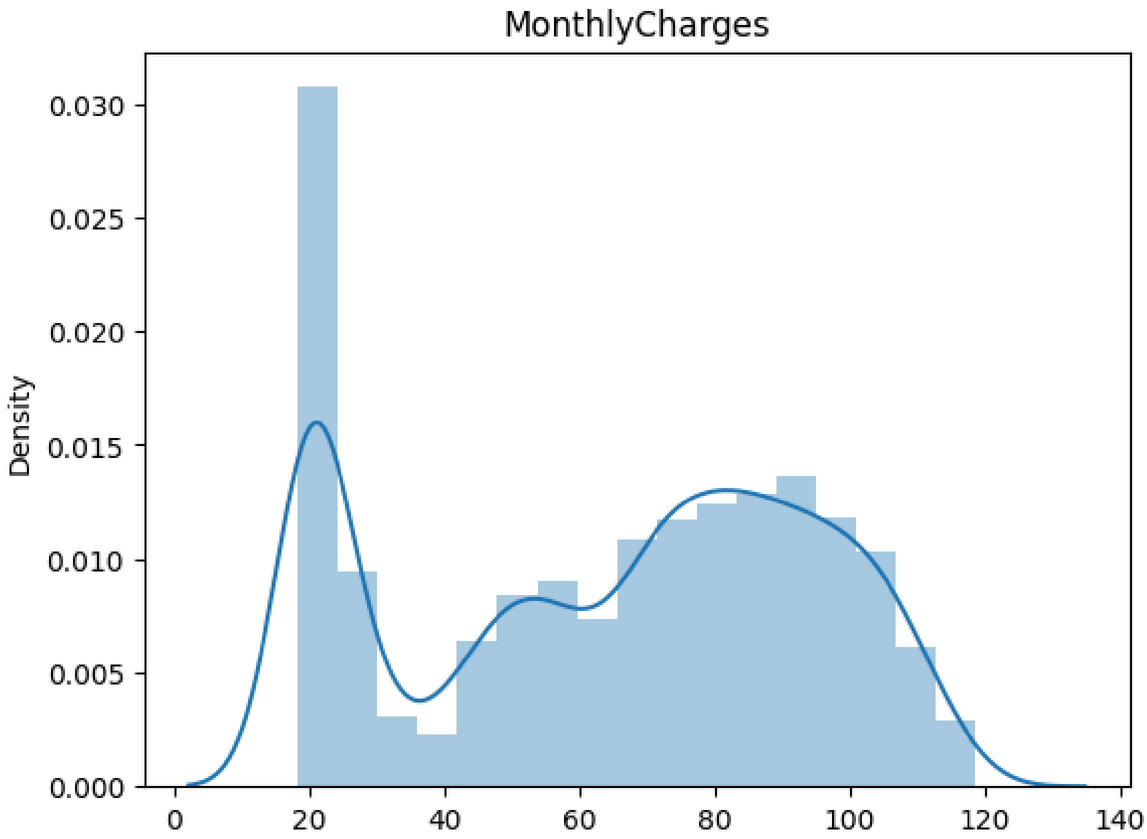
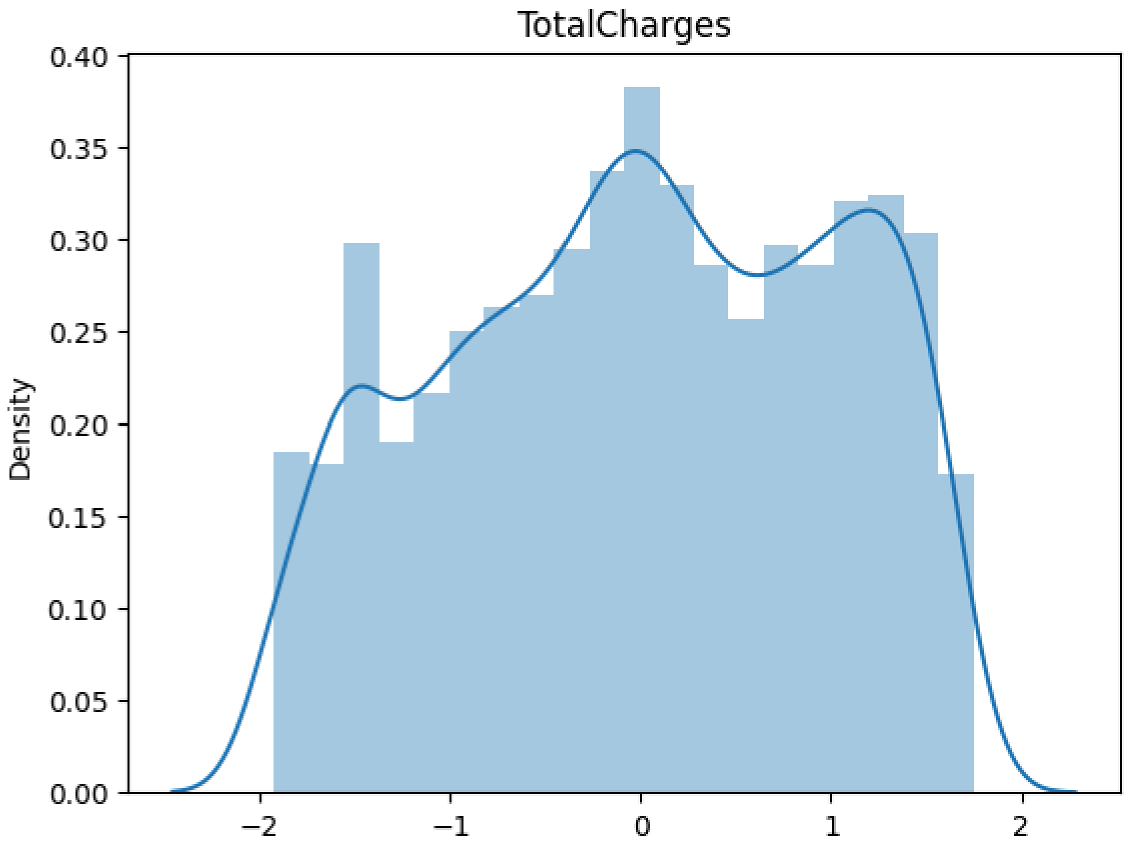




```
/tmp/ipykernel_2582/1145699575.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x=x_train[i])
```



In [120…
```
for i in numerical_columns:
  print(f'Skew of {i} is {x_train[i].skew()}')
```

```
Skew of tenure is 0.24154307949177942
Skew of MonthlyCharges is -0.2180067984814727
Skew of TotalCharges is -0.1453161207648667
```

In [120…
```
x_train.head()
```

Out[120…

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	In
0	Male	No	No	No	2	Yes	Yes	
1	Female	No	No	No	34	Yes	Yes	
2	Female	No	No	No	9	Yes	No	
3	Male	No	Yes	No	67	Yes	Yes	
4	Male	No	No	No	71	No	No phone service	

In [120…
```
categorical_columns = [i for i in x_train.select_dtypes('object').columns]
```

Encoding and Feature Scaling

In [120…

```
x_train.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5634 entries, 0 to 5633
Data columns (total 19 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            5634 non-null   object
 1   SeniorCitizen     5634 non-null   object
 2   Partner           5634 non-null   object
 3   Dependents        5634 non-null   object
 4   tenure            5634 non-null   int64
 5   PhoneService      5634 non-null   object
 6   MultipleLines     5634 non-null   object
 7   InternetService   5634 non-null   object
 8   OnlineSecurity    5634 non-null   object
 9   OnlineBackup      5634 non-null   object
 10  DeviceProtection  5634 non-null   object
 11  TechSupport       5634 non-null   object
 12  StreamingTV       5634 non-null   object
 13  StreamingMovies   5634 non-null   object
 14  Contract          5634 non-null   object
 15  PaperlessBilling  5634 non-null   object
 16  PaymentMethod     5634 non-null   object
 17  MonthlyCharges    5634 non-null   float64
 18  TotalCharges      5634 non-null   float64
dtypes: float64(2), int64(1), object(16)
memory usage: 836.4+ KB
```

In [120…

```
x_train['gender'].unique()
```

Out[120…

```
array(['Male', 'Female'], dtype=object)
```

In [120…

```
x_train['SeniorCitizen'].unique()
```

Out[120…

```
array(['No', 'Yes'], dtype=object)
```

In [120…

```
x_train['Partner'].unique()
```

Out[120…

```
array(['No', 'Yes'], dtype=object)
```

In [121…

```
x_train['Dependents'].unique()
```

Out[121…

```
array(['No', 'Yes'], dtype=object)
```

In [121…

```
x_train['PhoneService'].unique()
```

Out[121…

```
array(['Yes', 'No'], dtype=object)
```

In [121…

```
x_train['MultipleLines'].unique()
```

Out[121…

```
array(['Yes', 'No', 'No phone service'], dtype=object)
```

In [121…

```
x_train['InternetService'].unique()
```

Out[121…

```
array(['Fiber optic', 'DSL', 'No'], dtype=object)
```

In [121…
```
x_train['OnlineSecurity'].unique()
```
Out[121…
```
array(['No', 'Yes', 'No internet service'], dtype=object)
```
In [121…
```
x_train['OnlineBackup'].unique()
```
Out[121…
```
array(['Yes', 'No internet service', 'No'], dtype=object)
```
In [121…
```
x_train['DeviceProtection'].unique()
```
Out[121…
```
array(['No', 'Yes', 'No internet service'], dtype=object)
```
In [121…
```
x_train['TechSupport'].unique()
```
Out[121…
```
array(['No', 'Yes', 'No internet service'], dtype=object)
```
In [121…
```
x_train['StreamingMovies'].unique()
```
Out[121…
```
array(['No', 'Yes', 'No internet service'], dtype=object)
```
In [121…
```
x_train['Contract'].unique()
```
Out[121…
```
array(['Month-to-month', 'Two year', 'One year'], dtype=object)
```
In [122…
```
x_train['PaperlessBilling'].unique()
```
Out[122…
```
array(['Yes', 'No'], dtype=object)
```
In [122…
```
x_train['PaymentMethod'].unique()
```
Out[122…
```
array(['Electronic check', 'Mailed check', 'Bank transfer (automatic)',
       'Credit card (automatic)'], dtype=object)
```
In [122…
```
x_train['StreamingTV'].unique()
```
Out[122…
```
array(['No', 'Yes', 'No internet service'], dtype=object)
```
In [122…
```
ColumnTransformer_Preprocessor = ColumnTransformer([('Ordinal_encoding',OrdinalE
```
In [122…
```
ColumnTransformer_Preprocessor_y = ColumnTransformer([('Encoding',OrdinalEncoder
```
In [122…
```
pipelines = Pipeline([('Preprocessing',ColumnTransformer_Preprocessor)])
```
In [122…
```
pipelines_y = Pipeline([('Preprocessing_y',ColumnTransformer_Preprocessor_y)])
```
In [122…
```
x_train = pd.DataFrame(pipelines.fit_transform(x_train),columns=pipelines.named_
```
In [122…
```
x_test = pd.DataFrame(pipelines.transform(x_test),columns=pipelines.named_steps[
```
In [122…
```
y_train = pd.DataFrame(pipelines_y.fit_transform(y_train),columns=pipelines_y.na
```
In [123…
```
y_test = pd.DataFrame(pipelines_y.transform(y_test),columns=pipelines_y.named_st
```
In [123…
```
x_train.head()
```

Out[123...

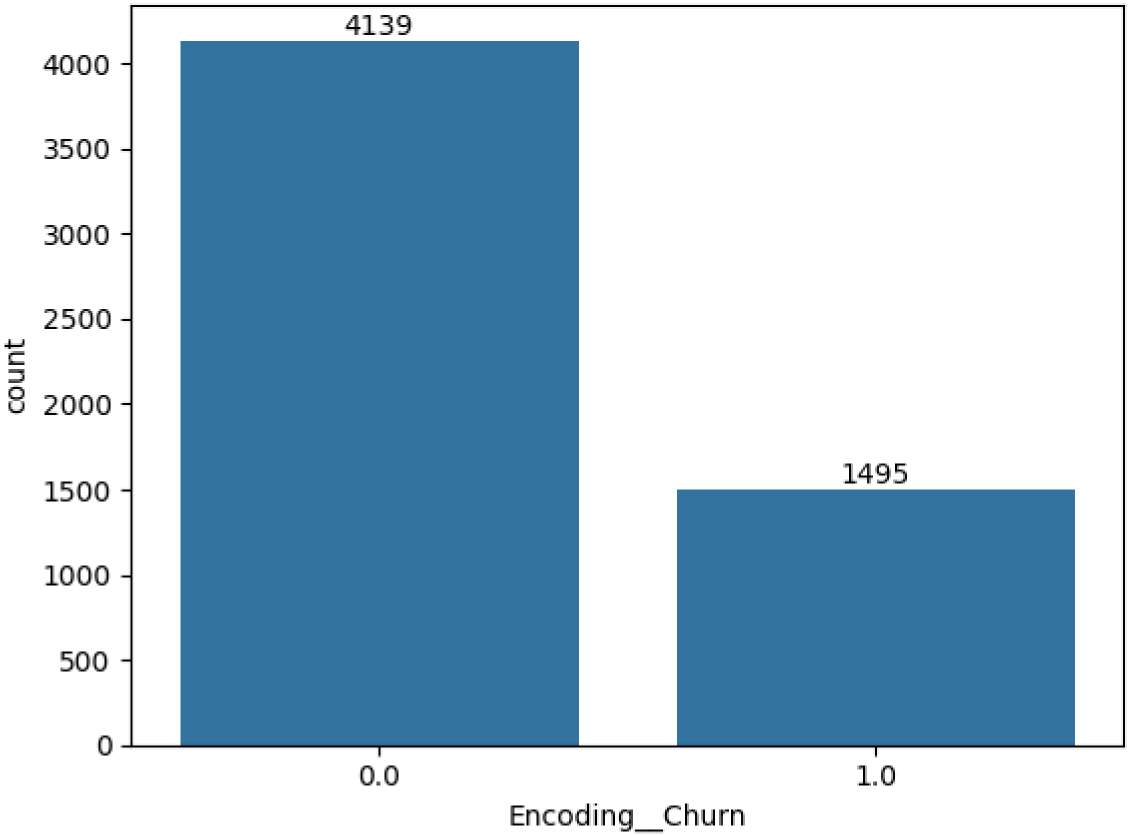
	Ordinal_encoding__gender	Ordinal_encoding__SeniorCitizen	Ordinal_encoding__Partner
0	1.0	0.0	0.0
1	0.0	0.0	0.0
2	0.0	0.0	0.0
3	1.0	0.0	1.0
4	1.0	0.0	0.0

5 rows × 40 columns

Imbalanced data handling

In [123...

```
plot = sns.countplot(x=y_train['Encoding__Churn'])
for i in plot.containers:
  plot.bar_label(i)
```



In [123...

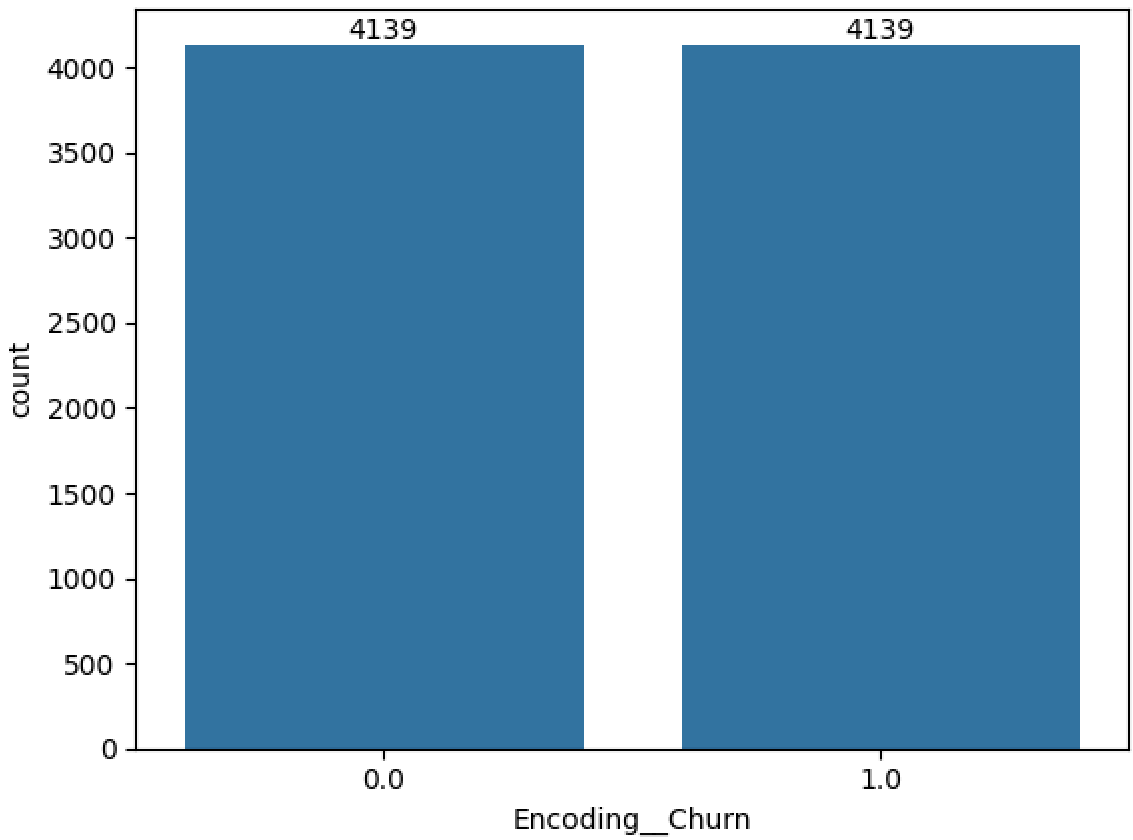
```
SMOTE_Churn = SMOTE()
```

In [123...

```
x_train, y_train = SMOTE_Churn.fit_resample(x_train,y_train)
```

In [123...

```
plot = sns.countplot(x=y_train['Encoding__Churn'])
for i in plot.containers:
  plot.bar_label(i)
```



Base model Creation

In [123…
```
RF_Base_Model = RandomForestClassifier(max_depth = 10, criterion='gini', max_lea
```

In [123…
```
RF_Base_Model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/base.py:1389: DataConversionWarni
ng: A column-vector y was passed when a 1d array was expected. Please change the
shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
```

Out[123…

▾ RandomForestClassifier

```
RandomForestClassifier(max_depth=10, max_leaf_nodes=20, n_estimators=20
0,
                       oob_score=True, random_state=21)
```

In [123…
```
RF_Base_Model.score(x_train,y_train)
```

Out[123… 0.7965692196182653

In [123…
```
RF_Base_Model.score(x_test,y_test)
```

Out[123… 0.7565649396735273

In [124…
```
y_train_pred = RF_Base_Model.predict(x_train)
```

In [124…
```
y_pred = RF_Base_Model.predict(x_test)
```

In [124…
```
precision_score(y_train,y_train_pred)
```
Out[124…
```
0.7875380651206372
```
In [124…
```
precision_score(y_test,y_pred)
```
Out[124…
```
0.5294117647058824
```
In [124…
```
recall_score(y_train,y_train_pred)
```
Out[124…
```
0.8122734960135298
```
In [124…
```
recall_score(y_test,y_pred)
```
Out[124…
```
0.7459893048128342
```

Hyper parameter tuning

In [124…
```
params = {'max_depth':[i for i in range(5,25)], 'max_leaf_nodes':[i for i in ran
```
In [124…
```
RandomSearch = RandomizedSearchCV(estimator = RandomForestClassifier(), param_di
```
In [124…
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RandomSearch.fit(x_train,y_train)
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```
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  return fit_method(estimator, *args, **kwargs)
```

Out[124…

RandomizedSearchCV

best_estimator_: RandomForestClassifier

RandomForestClassifier

In [124…

```
results = pd.DataFrame(RandomSearch.cv_results_)
```

In [125…

```
results.head()
```

Out[125…

	mean_fit_time	std_fit_time	mean_score_time	std_score_time	param_n_estimators	pa
0	1.819373	0.218241	0.063278	0.014209	450	
1	0.839177	0.293947	0.026872	0.000257	150	
2	1.259641	0.222977	0.046093	0.011335	250	
3	0.805039	0.023801	0.029458	0.000619	200	
4	1.638025	0.223832	0.058862	0.009897	350	

5 rows × 24 columns

In [125…

```
results[['param_n_estimators','param_max_leaf_nodes','param_max_depth','param_cr
```

Out[125…

	param_n_estimators	param_max_leaf_nodes	param_max_depth	param_criterion	me
7	300	5	7	entropy	
3	200	5	22	entropy	
0	450	5	9	gini	
22	350	7	17	entropy	
20	50	8	22	entropy	
18	200	6	9	entropy	
5	400	7	22	entropy	
2	250	7	24	entropy	
11	50	6	6	gini	
16	250	9	22	entropy	
8	250	5	20	gini	
6	450	9	20	entropy	
12	300	9	23	entropy	
17	100	9	21	entropy	
15	50	8	23	gini	
1	150	9	12	entropy	
19	150	6	23	gini	
24	450	9	5	gini	
10	300	9	13	gini	
14	150	9	6	entropy	
23	100	9	22	gini	
9	200	9	23	gini	
4	350	9	16	gini	
21	350	8	10	gini	
13	150	9	7	gini	

In [125…

```
best_params = RandomSearch.best_params_
```

In [125…

```
RF = RandomForestClassifier(**best_params,random_state=21)
```

In [125…

```
RF.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/base.py:1389: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
```

Out[125…

```
RandomForestClassifier
RandomForestClassifier(criterion='entropy', max_depth=7, max_leaf_nodes
=5,
                       n_estimators=300, random_state=21)
```

In [125…

```
y_pred = RF.predict(x_test)
```

In [125…

```
recall_score(y_pred,y_test)
```

Out[125…

```
0.49246231155778897
```

Treshold configuration

In [125…

```
y_prob = RF.predict_proba(x_test)[:,1]
```

In [125…

```
treshold = np.arange(0.1,0.9,0.1)
```

In [125…

```
for i in treshold:
  y_tresh_predict = (y_prob >= i).astype(int)
  print(f'Treshold : {i}')
  print(f'recall score :{recall_score(y_test,y_tresh_predict)}')
  print(f'recall score :{recall_score(y_test,y_tresh_predict)}')
  print()
```

```
Treshold : 0.1
recall score :1.0
recall score :1.0

Treshold : 0.2
recall score :0.9812834224598931
recall score :0.9812834224598931

Treshold : 0.30000000000000004
recall score :0.9572192513368984
recall score :0.9572192513368984

Treshold : 0.4
recall score :0.8636363636363636
recall score :0.8636363636363636

Treshold : 0.5
recall score :0.786096256684492
recall score :0.786096256684492

Treshold : 0.6
recall score :0.6550802139037433
recall score :0.6550802139037433

Treshold : 0.7000000000000001
recall score :0.36363636363636365
recall score :0.36363636363636365

Treshold : 0.8
recall score :0.0
recall score :0.0
```

Let the new treshold be 0.3

In [126...
```
y_train_prob = RF.predict_proba(x_train)[:,1]
```

In [126...
```
y_train_predict = (y_train_prob >= .3).astype(int)
```

In [126...
```
recall_score(y_train,y_train_predict)
```

Out[126...
```
0.9664170089393573
```

In [126...
```
y_predict = (y_prob >= .3).astype(int)
```

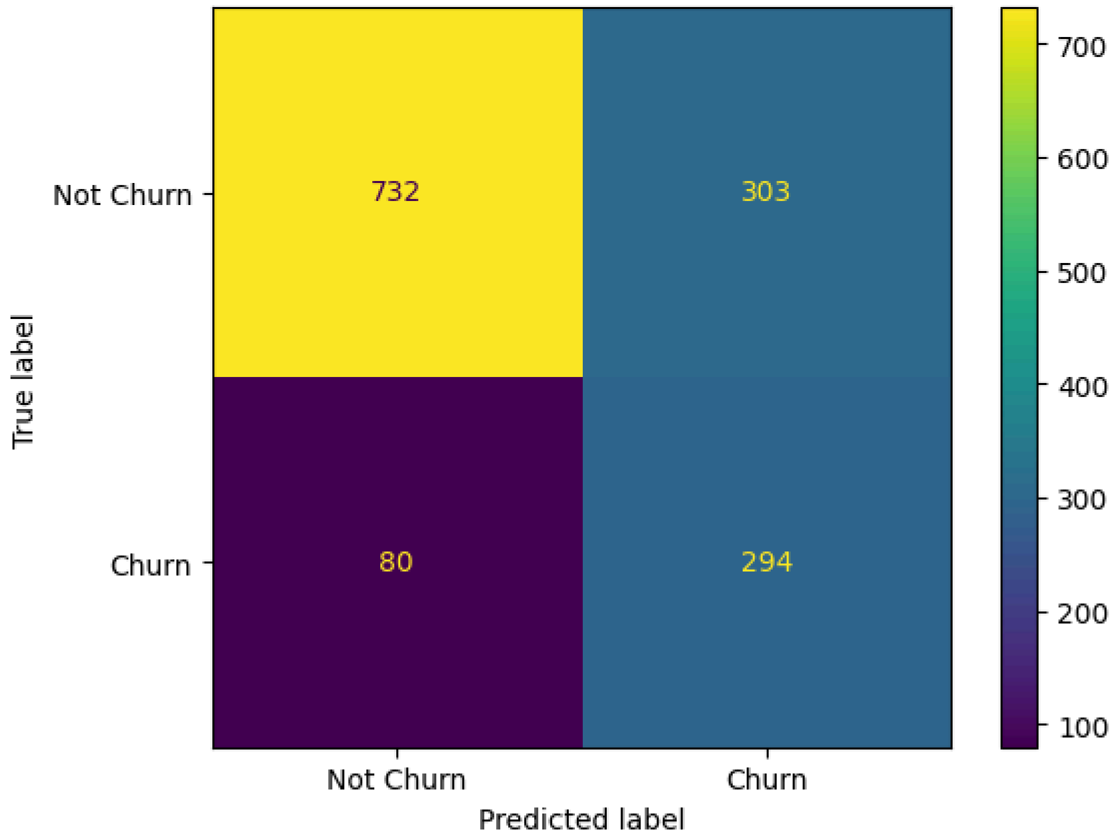
In [126...
```
recall_score(y_test,y_predict)
```

Out[126...
```
0.9572192513368984
```

Confusion Matrix

In [126...
```
cm = confusion_matrix(y_test,y_pred)
```

In [128...
```
ConfusionMatrixDisplay(cm,display_labels=['Not Churn','Churn']).plot()
plt.show()
```



After reducing the classification threshold to 0.3, the churn model became much better at identifying actual churn customers. The test recall increased to ~95.7%, which means the model is now able to correctly capture almost all churn cases. From the confusion matrix, the model correctly identified 294 churn customers and missed only 80 churn customers (false negatives), which is a strong outcome for a churn prediction problem where missing a churner is costly.

However, this improvement in recall comes with a trade-off: the model now produces more false positives — 303 non-churn customers were incorrectly predicted as churners. So the model is now highly aggressive in predicting churn, which is useful when the business goal is to minimize missed churn customers, even if it means targeting some customers who would not actually churn.

Overall conclusion: with a threshold of 0.3, the model is well-suited for churn prevention use cases where recall is the priority, because it captures most potential churners, but it sacrifices precision by increasing false alarms.

```
In [ ]:
```